Student Name: _____

Student ID# _____

	Check which tutorial for pick-up
T01 Monday mornings	
T02 Tuesday afternoons	

CHEMICAL ENGINEERING 2O04
Midterm #2

Duration 120 minutes (7:00pm – 9:00pm) **D.R. Latulippe**
McMaster University Term Test **Tuesday April 1st, 2025**

SPECIAL INSTRUCTIONS

1. The test includes 5 questions; ensure your copy of the test is complete.
2. Marks for each problem are given (**Total of 40 marks)**. Use this as a time-management guide; for example: $4\ marks\left(\frac{120\ minutes}{40\ marks}\right) \sim 12\ minutes$
3. The following materials are permitted during this examination:
   a) The Fox and McDonald's Introduction to Fluid Mechanics textbook (any edition and any version)
   b) Any printed notes or hand-written notes
   c) A tablet or computer in "airplane mode" with a document reader (e.g. Adobe Acrobat, Microsoft OneNote) open to review course material. **<u>Any student caught with a device not in airplane mode and/or with other applications opened will automatically receive a grade of 0 on the test.</u>**
   d) The McMaster Standard Calculator (Casio FX-991MS/MS+)
4. **The following materials are NOT permitted during this examination:**
   a) The internet.
   b) Any calculation software (e.g. Excel, MATLAB).
   c) Any applications to communicate with anyone.
5. Only tests written in pen will be considered if re-marking is requested (addition errors excepted).
6. **On page 10, we have given you a table with some useful conversion factors.**

For grading purposes only

Q1	Q2	Q3	Q4	Q5	TOTAL

QUESTION 1 (7 MARKS)

On slide 18 of the PDF_07 notes, we gave you the following equation for terminal velocity of a spherical particle:

$$v_t = \sqrt{\frac{4}{3}\frac{\left(\rho_{particle} - \rho_{fluid}\right) g D_p}{C_D \rho_{fluid}}}$$

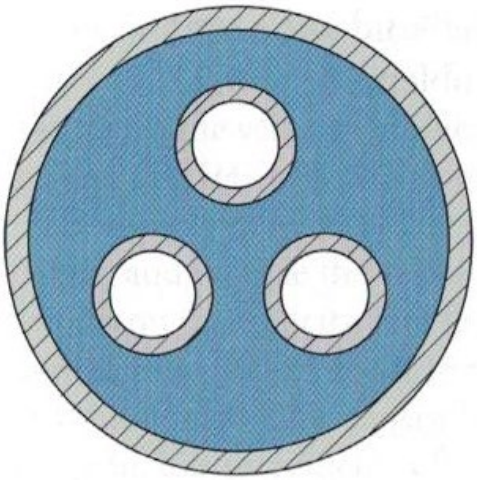
a) Prove that the equation is dimensionally consistent.
(1 mark)

b) Calculate the terminal velocity (m/s) of a grain of sand (diameter = 2 mm, density = 2300 kg/m^3) in ambient temperature air (i.e. 20 ^{o}C).
(6 marks)

QUESTION 2 (9 MARKS)

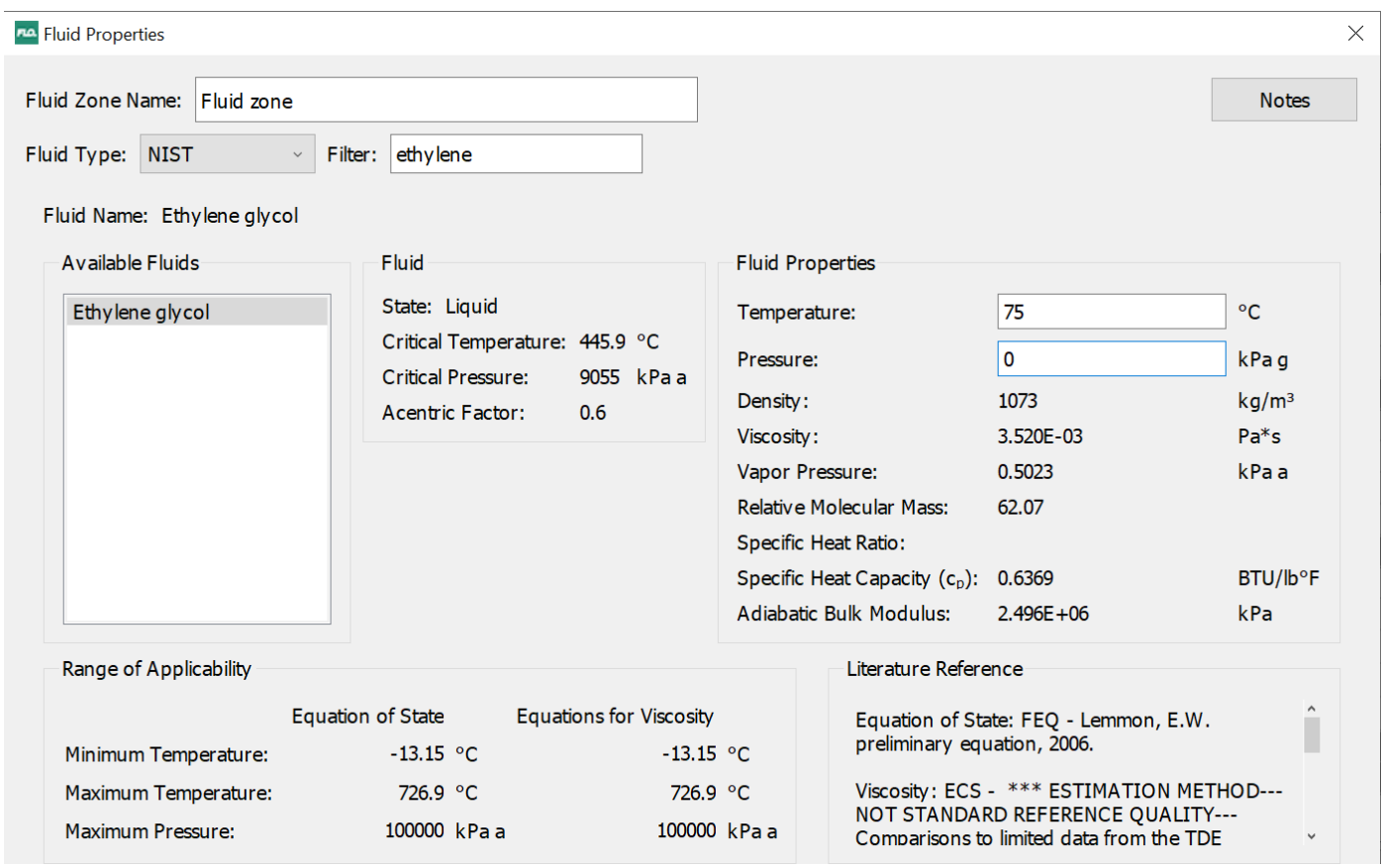
The accompanying figure is the cross-section of a heat-exchanger – the grey-hatched sections indicate the parts made of copper (chosen for its high thermal conductivity) and the solid blue-shaded 'shell' section is where the hotter fluid is flowing. The heat exchanger has the following dimensions:

- the outer shell has an internal diameter of 80 mm
- the three tubes on the inside have an external diameter of 25 mm
- the thickness of the shell and tubes are 1.5 mm



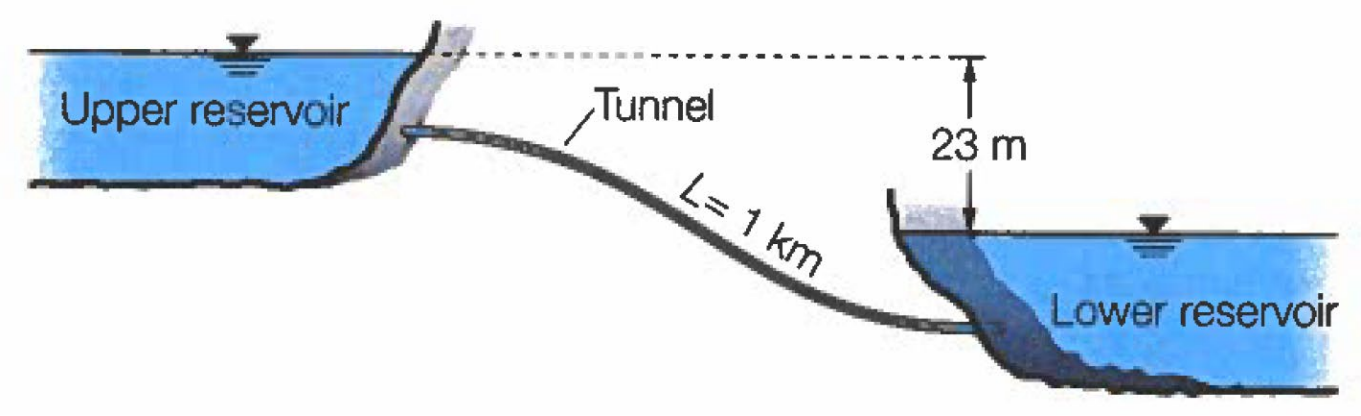
a) Calculate the Reynolds number when ethylene glycol at 75 ^{o}C (relevant properties according to PIPE-FLO are displayed at bottom of page 4) is the hotter fluid flowing at a volumetric rate of 950 L/min. **(5 marks)**

b) Calculate the pressure drop per unit length (kPa/m) in the shell section of a horizontal heat exchanger (i.e. no change in elevation).
(4 marks)



QUESTION 3 (9 MARKS)

Your engineering firm has been hired to review the design of a 1 kilometer long cylindrical tunnel that will connect an upper water reservoir to a lower water reservoir – the average water temperature (due to seasonal variations) is 10 ^{o}C. The tunnel will be lined with a polymeric coating with a roughness of 1.2 mm.



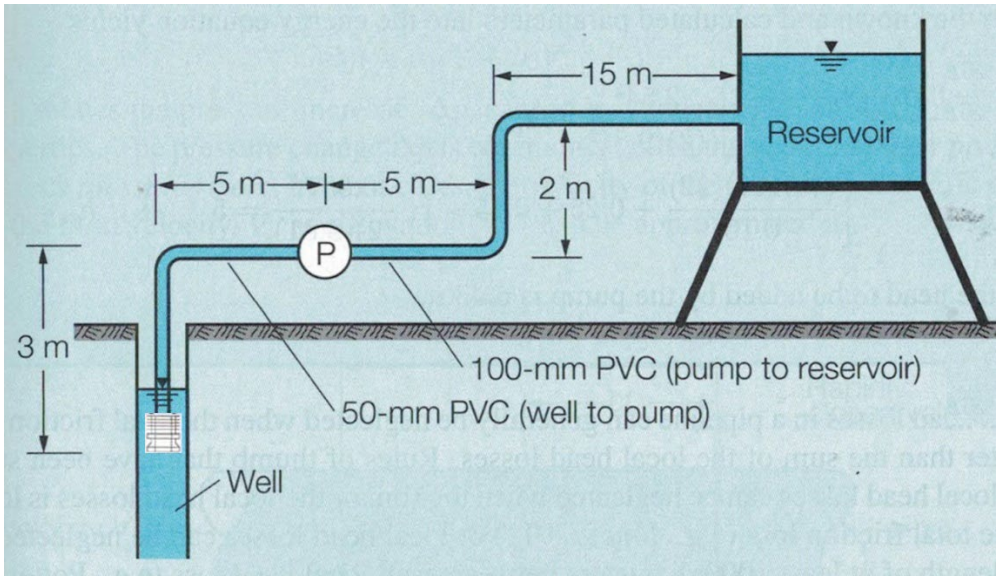
a) Write the simplified form of the Bernoulli equation that describes the transport of water from the upper reservoir to the lower reservoir. You can <u>ignore the entrance and exit losses.</u>
(2 marks)

b) To obtain a volumetric flow rate of 2 m^3/s inside the tunnel, a senior engineer at a consulting firm estimated that the tunnel diameter should be somewhere between 60 and 90 cm.
Prove/disprove the accuracy of that estimate.
(7 marks)

This page is intentionally left blank

QUESTION 4 (10 MARKS)

As shown in the figure below, a pump is being used to transfer 5 °C water from an underground well (in which the water level is constant due to it being continuously replenished by water flowing in from surrounding water table) to a large open-top reservoir.



a) Determine the elevation change (m) between the underground well and open-top reservoir given the following details:
   - When the head supplied by the pump is 9.25 m, the volumetric flow rate of water is 0.0048 m^3/s.
   - All the piping is PVC (essentially same as "smooth pipe"); *<u>the dimensions given on the figure are the actual internal diameters</u>*
   - There is a basket strainer (K = 1.3) on the inlet of the suction line
   - All the elbows are 90° glued regular (K = 0.31)

   (8 marks)

b) Using the information from part a), calculate the amount of power (W) reaching the fluid via the pump.
(2 marks)

QUESTION 5 (5 MARKS)

Calculate $NPSH_A$ (m) for the pump-piping system shown in Question 4.

Table G.2

Conversion Factors and Definitions

Fundamental Dimension	English Unit	Exact SI Value	Approximate SI Value
Length	1 in.	0.0254 m	—
Mass	1 lbm	0.453 592 37 kg	0.454 kg
Temperature	1°F	5/9 K	—

Definitions:

Acceleration of gravity:	$g = 9.8066\ \text{m/s}^2$ ($= 32.174\ \text{ft/s}^2$)
Energy:	Btu (British thermal unit) ≡ amount of energy required to raise the temperature of 1 lbm of water 1°F (1 Btu = 778.2 ft · lbf)
	kilocalorie ≡ amount of energy required to raise the temperature of 1 kg of water 1 K(1 kcal = 4187 J)
Length:	1 mile = 5280 ft; 1 nautical mile = 6076.1 ft = 1852 m (exact)
Power:	1 horsepower ≡ 550 ft · lbf/s
Pressure:	1 bar ≡ 10^5 Pa
Temperature:	degree Fahrenheit, $T_F = \frac{9}{5}T_C + 32$ (where T_C is degrees Celsius)
	degree Rankine, $T_R = T_F + 459.67$
	Kelvin, $T_K = T_C + 273.15$ (exact)
Viscosity:	1 Poise ≡ 0.1 kg/(m · s)
	1 Stoke ≡ 0.0001 m^2/s
Volume:	1 gal ≡ 231 in.^3 (1 ft^3 = 7.48 gal)

Useful Conversion Factors:

Length:	1 ft = 0.3048 m	Power:	1 hp = 745.7 W
	1 in. = 25.4 mm		1 ft · lbf/s = 1.356 W
Mass:	1 lbm = 0.4536 kg		1 Btu/hr = 0.2931 W
	1 slug = 14.59 kg	Area	1 ft^2 = 0.0929 m^2
Force:	1 lbf = 4.448 N		1 acre = 4047 m^2
	1 kgf = 9.807 N	Volume:	1 ft^3 = 0.02832 m^3
Velocity:	1 ft/s = 0.3048 m/s		1 gal (US) = 0.003785 m^3
	1 ft/s = 15/22 mph		1 gal (US) = 3.785 L
	1 mph = 0.447 m/s	Volume flow rate:	1 ft^3/s = 0.02832 m^3/s
Pressure:	1 psi = 6.895 kPa		1 gpm = 6.309×10^{-5} m^3/s
	1 lbf/ft^2 = 47.88 Pa	Viscosity (dynamic)	1 $\text{lbf} \cdot \text{s/ft}^2$ = 47.88 $\text{N} \cdot \text{s/m}^2$
	1 atm = 101.3 kPa		1 $\text{g/(cm} \cdot \text{s)}$ = 0.1 $\text{N} \cdot \text{s/m}^2$
	1 atm = 14.7 psi		1 Poise = 0.1 $\text{N} \cdot \text{s/m}^2$
	1 in. Hg = 3.386 kPa	Viscosity (kinematic)	1 ft^2/s = 0.0929 m^2/s
	1 mm Hg = 133.3 Pa		1 Stoke = 0.0001 m^2/s
Energy:	1 Btu = 1.055 kJ		
	1 ft · lbf = 1.356 J		
	1 cal = 4.187 J		

THE END